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MIDI CONTROLLERS

MIDI CONTROLLERS / How-to Articles

SysEx Documentation for External Applications



Work in progress

It is possible to interact with our MIDI controllers via SysEx messages, which allows you to update preset and bank parameters, or retrieve data from the controller like Preset Names etc.

Saving or Temporary Override

In most functions, there will be an option to save by sending 7F for a specific Opcode. If override is chosen instead (by using any other value), the values will just be written temporarily, and will revert back to the original values when the bank is changed.

For example, if you update the Preset Short Name from Empty to Preset 1 and do not save, when sending the SysEx string to update the Preset Short Name, the controller will display the Preset Name as Preset 1. However, when you change banks and go back to the same bank again, the Preset Name will show Empty.

SysEx Message Structure

For this API, Op1 is fixed at 0x70.

CODE

```
F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3
04 -- Device Model ID
00 -- Ignore
70 -- Opcode 1
```

```
00 -- Opcode 2
00 -- Opcode 3
00 -- Opcode 4
00 -- Opcode 5
00 -- Opcode 6
00 -- Opcode 7
xx -- Transaction ID
00 -- Ignore
00 -- Ignore
pp -- Payload Start
pp
pp
pp
yy -- CheckSum
F7 -- SysEx End
```

Message Payload

The message payload, which always starts at `Pointer 17`, contains the data for the intended function, and can be of **any length**. For example, if you want to update the Preset Name, the name will be in the Payload in ASCII code.

Transaction ID

The `Transaction ID` is used as an identifier for you to know which SysEx command the controller is responding to when it returns a SysEx message. For example, if you sent a request for a Preset Short Name with a Txn ID of `45`, the controller will send a response with a Txn ID of `45` as well, so you know it is responding to that specific request.

If there is no intention to use it, just leave it as `00`

CheckSum

The CheckSum is used to verify the message integrity. The calculation will be done on the sender's end, and then when the SysEx message is received by the Controller, it will calculate the CheckSum of the message and then compare it to the sender's CheckSum. If it does not match, the message is ignored.

The CheckSum can be calculated like this:

First, take the first value as F0 , which is the start of the SysEx message.

Next, XOR the value with the next value.

Continue this until you reach $n - 2$.

AND the value with 0x7F , which will give you a value of between 0 - 127.

Here is a sample code in C++:

CODE

```
int calculateChecksum(uint16_t len, uint8_t *ptr) {
    uint8_t cSum = *(ptr);
    for (int i = 1; i < len - 2; i++) {
        cSum ^= *(ptr + i);
    }
    cSum &= 0x7F;
    return cSum;
}
```

Return Codes

If there is any error in the received SysEx message, the Controller will return you a error message.

CODE

```
F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3
xx -- Device Model ID
00 -- Ignore
70 -- Opcode 1
7F -- Opcode 2
xx -- Ack Code
00 -- Opcode 4
00 -- Opcode 5
00 -- Opcode 6
00 -- Opcode 7
xx -- Transaction ID
```

```

00  -- Ignore
00  -- Ignore
yy  -- CheckSum
F7  -- SysEx End

```

where Ack Code is

CODE

```

SUCCESS                00
WRONG MODEL ID         01
WRONG CHECKSUM         02
WRONG PAYLOAD SIZE     03

```

Functions

This section describes the various functions as specified using Opcode 2.

[op2: 00h] Controller Functions

Using 00h for Op2 lets you access the Controller functions. The specific functions are selected using Op3.

[op3: 00h]Controller Bank Up

CODE

```
F0 00 21 24 id 00 70 00 00 00 00 00 00 00 00 cs F7
```

[op3: 01h]Controller Bank Down

CODE

```
F0 00 21 24 id 00 70 00 01 00 00 00 00 00 00 cs F7
```

[op3: 02h]Controller Toggle Page

CODE

```
F0 00 21 24 id 00 70 00 02 00 00 00 00 00 00 cs F7
```

[op2: 01h] Update Preset Short Name

- When updating names, the Payload length needs to match the name length limit for each device. For example, the short name length limit for the MC6MKII is 8 characters. When you want to update the name to "HELLO", which is 5 characters, you will need to add another 3 space characters (ASCII Code = 32) to make it 8 characters.

Please refer to the [Name Length Limits](#) in the Glossary.

CODE

```
F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3
xx -- Device Model ID
00 -- Ignore
70 -- Opcode 1
01 -- Opcode 2 - Update Preset Short Name
xx -- Opcode 3 - Preset Number (A = 0, B = 1 ...)
xx -- Opcode 4 - Whether to save to memory. 7F = Save.
00 -- Opcode 5
00 -- Opcode 6
00 -- Opcode 7
xx -- Transaction ID
00 -- Ignore
00 -- Ignore
pp -- Payload. Preset name in ASCII code
pp
....
pp
yy -- CheckSum
F7 -- SysEx End
```

[op2: 02h] Update Preset Toggle Name

CODE

```
F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3
xx -- Device Model ID
00 -- Ignore
70 -- Opcode 1
02 -- Opcode 2 - Update Preset Toggle Name
xx -- Opcode 3 - Preset Number (A = 0, B = 1 ...)
xx -- Opcode 4 - Whether to save to memory. 7F = Save.
00 -- Opcode 5
00 -- Opcode 6
00 -- Opcode 7
xx -- Transaction ID
00 -- Ignore
00 -- Ignore
pp -- Payload. Preset name in ASCII code
pp
....
pp
yy -- CheckSum
F7 -- SysEx End
```

[op2: 03h] Update Preset Long Name

CODE

```
F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3
xx -- Device Model ID
00 -- Ignore
70 -- Opcode 1
03 -- Opcode 2 - Update Preset Long Name
xx -- Opcode 3 - Preset Number (A = 0, B = 1 ...)
xx -- Opcode 4 - Whether to save to memory. 7F = Save.
00 -- Opcode 5
```

```

00 -- Opcode 6
00 -- Opcode 7
xx -- Transaction ID
00 -- Ignore
00 -- Ignore
pp -- Payload. Preset name in ASCII code
pp
....
pp
yy -- CheckSum
F7 -- SysEx End

```

[op2: 04h] Update Preset Message

CODE

```

F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3
xx -- Device Model ID
00 -- Ignore
70 -- Opcode 1 - 0x70
04 -- Opcode 2 - Update Preset Message
xx -- Opcode 3 - Preset Number (A = 0, B = 1 ...)
xx -- Opcode 4 - Message Number (0-15, or 0x00 - 0x0F)
00 -- Opcode 5 - Message Type (See MESSAGE TYPES)
00 -- Opcode 6 - Whether to save to memory. 7F = Save.
00 -- Opcode 7
xx -- Transaction ID
00 -- Ignore
00 -- Ignore
pp -- Payload
pp
....
pp
yy -- CheckSum
F7 -- SysEx End

```

Message Types

The required payload will depend on the type of message selected with Opcode 5

[op5: 01h] Program Change

Required Payload

CODE

```

pp -- Action Type (See ACTION TYPES)
pp -- Toggle Type
pp -- PC Number
pp -- MIDI Channel

```

[op5: 02h] Control Change

Required Payload

CODE

```

pp -- Action Type (See ACTION TYPES)
pp -- Toggle Type
pp -- CC Number
pp -- CC Value
pp -- MIDI Channel

```

[op2: 05h] Other Data Preset

CODE

```

F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3
xx -- Device Model ID
00 -- Ignore
70 -- Opcode 1 - 0x70
05 -- Opcode 2 - Update Preset Other Data
xx -- Opcode 3 - Preset Number (A = 0, B = 1 ...)
xx -- Opcode 4 - Message Number (0-15, or 0x00 - 0x0F)
00 -- Opcode 5 - Message Type (See MESSAGE TYPES)
00 -- Opcode 6 - Whether to save to memory. 7F = Save.
00 -- Opcode 7

```

```

00 -- Opcode 1
xx -- Transaction ID
00 -- Ignore
00 -- Ignore
pp -- Preset Toggle: 7F = ON, 00 = OFF, Other = IGNORE
pp -- Preset Blink: 7F = ON, 00 = OFF, Other = IGNORE
pp -- Preset Message Scroll Mode: 7F = ON, 00 = OFF, Other = IGNORE
pp -- Preset Global Toggle Group: 00 = Independent, 01-10 = Groups 1 to 16, Other = I

[Ignore this if not using the PRO model]
pp -- Pos 1 LED Color      (PRO only) F7 = IGNORE
pp -- Pos 2 LED Color      (PRO only) F7 = IGNORE
pp -- Shift LED Color      (PRO only) F7 = IGNORE
pp -- Pos 1 Text Color      (PRO only) F7 = IGNORE
pp -- Pos 2 Text Color      (PRO only) F7 = IGNORE
pp -- Shift Text Color      (PRO only) F7 = IGNORE
pp -- Pos 1 Background Color (PRO only) F7 = IGNORE
pp -- Pos 2 Background Color (PRO only) F7 = IGNORE
pp -- Shift Background Color (PRO only) F7 = IGNORE

yy -- CheckSum
F7 -- SysEx End

```

[op2: 10h] Update Current Bank Name

CODE

```

F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3
xx -- Device Model ID
00 -- Ignore
70 -- Opcode 1 - 0x70
10 -- Opcode 2 - Update Current Bank Name
00 -- Opcode 3
00 -- Opcode 4 - Whether to save to memory. 7F = Save.
00 -- Opcode 5
00 -- Opcode 6
00 -- Opcode 7
xx -- Transaction ID
00 -- Ignore
00 -- Ignore

```

```
ww -- ignore
pp -- Payload. Bank name in ASCII code
pp
....
pp
yy -- CheckSum
F7 -- SysEx End
```

[op2: 11h] Display message on LCD

Displays a custom message of up to 20 characters in the controller's LCD.

CODE

```
F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3
xx -- Device Model ID
00 -- Ignore
70 -- Opcode 1 - 0x70
11 -- Opcode 2 - Display message on LCD
00 -- Opcode 3
00 -- Opcode 4 - Duration to display message. Increments of 100ms. Value of 10 = 1000
00 -- Opcode 5
00 -- Opcode 6
00 -- Opcode 7
00 -- Ignore
00 -- Ignore
00 -- Ignore
pp -- Payload. Message in ASCII code up to 20 characters
pp
....
pp
yy -- CheckSum
F7 -- SysEx End
```

[op2: 21h] Get Preset Short Name

CODE

```
F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3
xx -- Device Model ID
00 -- Ignore
70 -- Opcode 1 - 0x70
21 -- Opcode 2 - Get Preset Short Name
00 -- Opcode 3 - Preset Number (A = 0, B = 1 ...)
00 -- Opcode 4
00 -- Opcode 5
00 -- Opcode 6
00 -- Opcode 7
xx -- Transaction ID
00 -- Ignore
00 -- Ignore
yy -- CheckSum
F7 -- SysEx End
```

Return SysEx

CODE

```
F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3
xx -- Device Model ID
00 --
70 -- Opcode 1
21 -- Opcode 2
xx -- Opcode 3 - Preset Number (A = 0, B = 1 ...)
xx -- Opcode 4 - Preset Name Length / Payload Size
00 -- Opcode 5
00 -- Opcode 6
00 -- Opcode 7
xx -- Transaction ID
00 -- Ignore
00 -- Ignore
pp -- Payload
pp
....
pp
```

```
yy  -- CheckSum
F7  -- SysEx End
```

[op2: 22h] Get Preset Toggle Name

CODE

```
F0  -- SysEx Start
00  -- Manufacturer ID 1
21  -- Manufacturer ID 2
24  -- Manufacturer ID 3
xx  -- Device Model ID
00  -- Ignore
70  -- Opcode 1 - 0x70
22  -- Opcode 2 - Get Preset Toggle Name
00  -- Opcode 3 - Preset Number (A = 0, B = 1 ...)
00  -- Opcode 4
00  -- Opcode 5
00  -- Opcode 6
00  -- Opcode 7
xx  -- Transaction ID
00  -- Ignore
00  -- Ignore
yy  -- CheckSum
F7  -- SysEx End
```

Return SysEx

CODE

```
F0  -- SysEx Start
00  -- Manufacturer ID 1
21  -- Manufacturer ID 2
24  -- Manufacturer ID 3
xx  -- Device Model ID
00  --
70  -- Opcode 1
22  -- Opcode 2
xx  -- Opcode 3 - Preset Number (A = 0, B = 1 ...)
xx  -- Opcode 4 - Preset Name Length / Payload Size
00  -- Opcode 5
00  -- Opcode 6
```

```

00 -- Opcode 7
xx -- Transaction ID
00 -- Ignore
00 -- Ignore
pp -- Payload
pp
....
pp
yy -- CheckSum
F7 -- SysEx End

```

[op2: 23h] Get Preset Long Name

CODE

```

F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3
xx -- Device Model ID
00 -- Ignore
70 -- Opcode 1 - 0x70
23 -- Opcode 2 - Get Preset Long Name
00 -- Opcode 3 - Preset Number (A = 0, B = 1 ...)
00 -- Opcode 4
00 -- Opcode 5
00 -- Opcode 6
00 -- Opcode 7
xx -- Transaction ID
00 -- Ignore
00 -- Ignore
yy -- CheckSum
F7 -- SysEx End

```

Return SysEx

CODE

```

F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3

```

```
24 -- Manufacturer ID 3
xx -- Device Model ID
00 --
70 -- Opcode 1
23 -- Opcode 2
xx -- Opcode 3 - Preset Number (A = 0, B = 1 ...)
xx -- Opcode 4 - Preset Name Length / Payload Size
00 -- Opcode 5
00 -- Opcode 6
00 -- Opcode 7
xx -- Transaction ID
00 -- Ignore
00 -- Ignore
pp -- Payload
pp
....
pp
yy -- CheckSum
F7 -- SysEx End
```

[op2: 30h] Get Current Bank Name

CODE

```
F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3
xx -- Device Model ID
00 -- Ignore
70 -- Opcode 1 - 0x70
30 -- Opcode 2 - Get Current Bank Name
00 -- Opcode 3
00 -- Opcode 4
00 -- Opcode 5
00 -- Opcode 6
00 -- Opcode 7
xx -- Transaction ID
00 -- Ignore
00 -- Ignore
yy -- CheckSum
F7 -- SysEx End
```

Return SysEx

CODE

```

F0  -- SysEx Start
00  -- Manufacturer ID 1
21  -- Manufacturer ID 2
24  -- Manufacturer ID 3
xx  -- Device Model ID
00  --
70  -- Opcode 1
21  -- Opcode 2
00  -- Opcode 3
xx  -- Opcode 4 - Bank Name Length / Payload Size
00  -- Opcode 5
00  -- Opcode 6
00  -- Opcode 7
xx  -- Transaction ID
00  -- Ignore
00  -- Ignore
pp  -- Payload
pp
....
pp
yy  -- CheckSum
F7  -- SysEx End

```

[op2: 31h] Get Toggle State of all Presets in current Bank

CODE

```

F0  -- SysEx Start
00  -- Manufacturer ID 1
21  -- Manufacturer ID 2
24  -- Manufacturer ID 3
xx  -- Device Model ID
00  -- Ignore
70  -- Opcode 1 - 0x70
31  -- Opcode 2 - Get toggle states
00  -- Opcode 3
00  -- Opcode 4
--  --

```

```

00 -- Opcode 5
00 -- Opcode 6
00 -- Opcode 7
xx -- Transaction ID
00 -- Ignore
00 -- Ignore
yy -- CheckSum
F7 -- SysEx End

```

Return SysEx

CODE

```

F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3
xx -- Device Model ID
00 --
70 -- Opcode 1
31 -- Opcode 2
00 -- Opcode 3
xx -- Opcode 4 - Total number of presets / Payload Size
00 -- Opcode 5
00 -- Opcode 6
00 -- Opcode 7
xx -- Transaction ID
00 -- Ignore
00 -- Ignore
pp -- Payload
pp
....
pp
yy -- CheckSum
F7 -- SysEx End

```

where each Payload value contains the toggle state of each preset. 7f = Toggled

Payload

CODE

```

00 -- Preset A, not toggled
AA -- Preset R not toggled

```

```

00 -- Preset B, not toggled
7F -- Preset C, toggled
...
00 -- Preset n, not toggled

```

[op2: 32h] Get Controller Information

Retrieve information like Model ID, firmware version etc.

CODE

```

F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3
xx -- Device Model ID
00 -- Ignore
70 -- Opcode 1 - 0x70
32 -- Opcode 2 - Get Controller Information
00 -- Opcode 3
00 -- Opcode 4
00 -- Opcode 5
00 -- Opcode 6
00 -- Opcode 7
xx -- Transaction ID
00 -- Ignore
00 -- Ignore
yy -- CheckSum
F7 -- SysEx End

```

Return SysEx

CODE

```

F0 -- SysEx Start
00 -- Manufacturer ID 1
21 -- Manufacturer ID 2
24 -- Manufacturer ID 3
xx -- Device Model ID
00 --
70 -- Opcode 1
32 -- Opcode 2
00 -- Opcode 3

```

```
00 -- Opcode 3
09 -- Opcode 4 - Payload Size
00 -- Opcode 5
00 -- Opcode 6
00 -- Opcode 7
xx -- Transaction ID
00 -- Ignore
00 -- Ignore
pp -- Model ID
pp -- Firmware Version 1
pp -- Firmware Version 2
pp -- Firmware Version 3
pp -- Firmware Version 4
pp -- Total Messages per Preset
pp -- Preset Name Size
pp -- Preset Long Name Size
pp -- Bank Name Size
yy -- CheckSum
F7 -- SysEx End
```

Glossary

Device Model ID

CODE

MC6	0x03
MC8	0x04
MC3	0x05
MC6PRO	0x06
MC8PRO	0x08

Action Types

This section specifies the Action IDs required when programming Preset Messages

CODE

NOTHING	0x00
PRESS	0x01
RELEASE	0x02

LONG PRESS	0x03
LONG PRESS RELEASE	0x04
DOUBLE TAP	0x05
DOUBLE TAP RELEASE	0x06
DOUBLE TAP LONG	0x07
DOUBLE TAP LONG RELEASE	0x08
RELEASE ALL	0x09
LONG PRESS SCROLL	0x0A
ON DISENGAGE	0x0B
ON FIRST ENGAGE	0x0C

Message Types

This section specifies the Message Type IDs required when programming Preset Messages

CODE

NOTHING	0x00
PC MESSAGE	0x01
CC MESSAGE	0x02

Preset Message Toggle Types

This section specifies the Message Type IDs required when programming Preset Messages

CODE

POS 1	0x00
POS 2	0x01
POS BOTH	0x02
SHIFT	0x03

Name Length Limits

MC3

CODE

Short Name	10
------------	----

Toggle Name	10
Long Name	16
Bank Name	16

MC6MKII

CODE

Short Name	8
Toggle Name	8
Long Name	24
Bank Name	24

MC68

CODE

Short Name	10
Toggle Name	10
Long Name	24
Bank Name	24

MC6 PRO

CODE

Short Name	32
Toggle Name	32
Long Name	32
Bank Name	32

MC8 PRO

CODE

Short Name	32
Toggle Name	32
Long Name	32
Bank Name	32

Color Chart



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